

Model Card — Sarcasm Detector for News Headlines

1. Model details

Field	Description
Model name	SarcasmDetect-v1
Task	Binary classification (sarcastic / real) of short English news headlines
Model	Fine-tuned distilbert-base-uncased (2 epochs, batch size 32, learning rate 2e-5, max 48 tokens). Baseline: TF-IDF (unigrams + bigrams, 10,000 features) + Logistic Regression (C = 3, chosen by 5-fold cross-validation on training data only)
Training data	Misra's News Headlines Dataset for Sarcasm Detection v2 (public): 28,503 unique headlines, 47.5% sarcastic (sarcastic = The Onion, real = HuffPost). 22,802 used for training
Test data	5,701 held-out headlines (random 80/20 split, random seed 42): 2,653 sarcastic, 3,048 real
Preprocessing	TF-IDF models: lowercasing, punctuation stripping, lemmatization (words over 3 letters), English stop words removed. DistilBERT: raw headline text with its own tokenizer
Libraries	Python 3 (Google Colab, T4 GPU), scikit-learn, Hugging Face Transformers, NLTK, LIME
Developed by	Sukriti Singh (PRN 25030422121), SAIL — NLP course, October 2026

2. Intended use

- **Intended:** Classroom demonstration of an NLP pipeline with a bias and explainability audit on English news headlines.
- **Out of scope:** Moderating or judging real people's posts; decisions about individuals; everyday or conversational sarcasm (social media, reviews, chat); other languages or code-mixed text (e.g., Hinglish); medical, legal, or safety-critical content.

3. Overall performance

The model reaches 92.0% accuracy and 0.919 macro-F1 (0.913 F1 on the sarcastic class) on the 5,701-headline test set, beating a majority-class baseline (53.5%) by 38.5 points and the TF-IDF + Logistic Regression baseline (79.7%) by 12.3 points. Precision, recall and F1 below are for the sarcastic class.

Model	Accuracy	Precision	Recall	F1	AUC
Majority-class baseline (always “real”)	53.5%	—	—	—	—
TF-IDF + Logistic Regression	79.7%	0.790	0.769	0.779	0.878
SarcasmDetect-v1 (DistilBERT)	92.0%	0.923	0.903	0.913	0.978

4. Performance across text subgroups

Key audit finding: On headline-style text the model is steady across subgroups (91–95% accuracy), but on a 24-text stress test it caught 4 of 4 Onion-style sarcastic texts and 0 of 8 everyday sarcastic texts. Sarcasm recall is also 10 points lower on first-person headlines (80.4% vs 90.5%), though with only 46 sarcastic examples this gap could be noise.

Subgroup / style	How it was defined	Test size (n)	Accuracy	Macro-F1
Short headlines	Fewer than 10 words	2,489	91.0%	0.910
Long headlines	10 words or more	3,212	92.7%	0.926
Third-person voice	No I / me / my / we in the headline	5,393	91.8%	0.918
First-person voice	Contains I, me, my, mine, we, our, us (closer to everyday speech)	308	94.8%	0.896
Onion-style sarcasm	Hand-written stress test; all sarcastic, so accuracy = recall	4	100%	—
Everyday sarcasm	Hand-written stress test: 4 dry + 4 with markers (“ <i>yeah right</i> ”, “ <i>sure, because...</i> ”); accuracy = recall	8	0%	—

The model was built on two news outlets’ headlines, so it is least reliable exactly where sarcasm is most everyday: casual, first-person, non-headline text.

5. Explainability check

Method 1 — Top features (Logistic Regression coefficients). The strongest words are Onion newsroom templates (sarcastic) and HuffPost topics (real), not irony itself, which suggests the baseline learned the outlet’s style rather than sarcasm.

Top sarcastic features (weight)	Top real features (weight)
area (+7.5)	queer (-4.6)
nation (+7.4)	trump (-4.4)
report (+6.5)	allegedly (-4.3)
man (+6.4)	trans (-4.2)
clearly (+5.7)	donald (-4.2)

Method 2 — LIME on a misclassified text (fine-tuned DistilBERT).

- Text: “*I absolutely love being stuck in traffic*” (hand-written stress test) — true label: sarcastic; predicted: **real** ($P(\text{sarcastic}) = 0.00$).
- LIME shows *love* (-0.28) and *I* (-0.18) pushing towards real, while *being* (+0.28) and *stuck* (+0.22) push towards sarcastic but not enough to change the answer.
- **Interpretation:** the sincere-sounding words “*I*” and “*love*” act as real-news cues and nothing looks like an Onion template, so dry sarcasm is missed. This is consistent with the 0 of 8 everyday-sarcasm result in Section 4.
- **Global check:** over 80 test headlines, DistilBERT’s strongest sarcastic cues are *area*, *man*, *old*, *report*, *time* and its strongest real cues are *is*, *will*, *says*, *the*, *why*. Removing the 49 top-weight words from the baseline changes its accuracy by -3.5 pts, versus -0.2 pts for 49 random words.

Logistic Regression weights are read from the notebook plot (about ± 0.1). LIME weights are approximate and vary slightly between runs.

6. Limitations and ethical considerations

- **Main limitation:** The model mostly recognises The Onion’s headline style, not sarcasm in general. It caught 0 of 8 everyday sarcastic texts and was confidently wrong ($P(\text{sarcastic})$ under 0.07 for 6 of those 8). It should not be used on conversational text.
- **Data bias:** Only two US-centred news outlets. The baseline’s strongest “real” cues include topic words such as *queer*, *trans*, *transgender*, *muslim*, so topic can override tone for texts about these groups. This topic shortcut was not tested for DistilBERT.
- **Label noise:** Labels come from the publication, not from people judging sarcasm: The Onion is satire (not always sarcasm) and HuffPost is assumed never to be sarcastic.
- **Small audit sample:** The stress test has only 24 texts (12 sarcastic). First-person headlines have 46 sarcastic examples and question headlines 23 (sarcasm recall 52%), too few to trust. With 20 subgroup comparisons, about one “real gap” may be chance. Results come from a single training run.

7. Recommendations

- Train and test on human-labelled sarcasm (e.g., iSarcasm or Reddit comments) and add everyday sarcasm, so the label means sarcasm rather than outlet.
- Enlarge the stress test (hundreds of texts per category) and repeat training over several random seeds before drawing conclusions.
- Do not rely on a confidence threshold to catch errors, because the model was confidently wrong on everyday sarcasm; send text that is not headline-style to a human reviewer instead.